

LESSON 6-5 • SESSION 2 • PRACTICE • FORM B

Evaluating Algebraic Expressions

Learning target – I can evaluate an algebraic expression by substituting a value for the variable. • 6.EE.A.2c

REMEMBER

- To **evaluate**, substitute the value, then use the **order of operations**.
- $4x$ means $4 \times x$. When $x = 3$, $4x = 12$, not 43.
- Powers come before multiplying: $2n^2$ with $n = 5$ is $2 \times 25 = 50$.

WORKED EXAMPLEEvaluate $2n^2 + 3$ when $n = 5$.**Step 1** Substitute: $2 \times 5^2 + 3$.**Step 2** Power first: $5^2 = 25$, so $2 \times 25 + 3$.**Step 3** Multiply, then add: $50 + 3 = 53$.**Answer 53****WORDS** substitute (*sustituir*) • evaluate (*evaluar*) • variable (*variable*)**PRACTICE***show your work • label units*

1. Evaluate each expression when
- $c = 8$
- .

a. $3c =$ _____

b. $c + 19 =$ _____

c. $60 - 5c =$ _____

d. $c^2 - 10 =$ _____

e. $c \div 4 + 6 =$ _____

f. $3(c - 2) =$ _____

- 2.
- Art Studio.**
- An art studio charges
- $6h + 7$
- dollars for
- h
- hours of studio time.

a. Cost for 2 hours: _____ dollars

b. Cost for 5 hours: _____ dollars

c. How much more is 5 hours than 2 hours? _____ dollars

d. What does the 7 do to every cost? _____

- 3.
- Bike Rental.**
- The expression
- $8r + 15$
- gives the cost in dollars for
- r
- hours. Find the cost for 6 hours.

Hint Multiply first, then add 15.*Show your work.***Answer** _____ dollars

- 4.
- Power Inside.**
- Evaluate
- $3n^2 + 7$
- when
- $n = 4$
- .

Hint Square 4 before you multiply by 3.*Show your work.***Value** _____

- 5.
- Ken's Work.**
- Ken evaluated
- $6 + 3p$
- when
- $p = 4$
- and got 36. Justify or correct his work.

Hint After you substitute, which operation comes first?*Show your work.***Value** _____

6. **Filling Tank.** The expression $12 + 5h$ gives the liters of water in a tank after h hours.

Hint Substitute each value on its own.

- a. Liters when $h = 3$: _____ L
- b. Liters when $h = 9$: _____ L
- c. What happens as h grows? _____

Show your work.

7. **Team Points.** A league uses $3w + 2t$ for points, where w is wins and t is ties.

Hint Substitute both values, then add the products.

- a. Points for 6 wins and 2 ties: _____ points
- b. Points for 7 wins and 0 ties: _____ points

Show your work.

8. **Phone Plan.** A plan costs $12g + 20$ dollars, where g is the number of gigabytes used.

- a. Cost for 4 gigabytes: _____ dollars
- b. Cost for 8 gigabytes: _____ dollars
- c. The gigabytes doubled. Why didn't the cost double? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Eli's Work.** Eli wrote "12 less than a number y " as $12 - y$. Explain his mistake. Then write the correct expression and evaluate it when $y = 30$.

TEST PRACTICE

choose the best answer

10. Find the value of $5x$ when $x = 6$.

- A** 56 **B** 11 **C** 30 **D** 1

11. Find the value of $20 - 3k$ when $k = 5$.

- A** 85 **B** 5 **C** 2 **D** 12

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

18 I	55 R	63 A	22 B	116 E	36 Q	151 Z	27 A	37 V	68 L
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#2b	#3	#4	#5	#6a	#7a	#8a	#8b

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____